

Murali Karthick Muthuvel

murali47karthick@gmail.com | Github profile | Linkedin profile | Kochi, Kerala, India

PROFILE

Computer Science undergraduate at IIT Palakkad specializing in **Machine Learning, Deep Learning, Large Language Models (LLMs), NLP, and Explainable AI**. Research experience spans multi-agent LLM systems, model interpretability, quantum-enhanced graph learning, and applied deep learning for scientific and medical domains. Proficient in Python, PyTorch, TensorFlow, LangGraph and LangChain.

EDUCATION

Indian Institute of Technology Palakkad
B.Tech. in Computer Science and Engineering

2023 – 2027
CGPA: 9.38/10

TOEFL iBT: 110/120 (CEFR C1)
GRE: 327.5/346 (Quant + Verbal)

RESEARCH & TECHNICAL EXPERIENCE

Tata Consultancy Services Research
Research Intern

May – Jul 2026
Pune, India

- Built a multi-agent **LLM** system using **LangGraph** and **GPT-4o** for automated functional test-case generation from software requirements.
- Engineered Planner, Generator, Verifier, Judge and Supervisor agents for iterative generation, validation and refinement.
- Integrated Python, LangChain and automated evaluation pipelines for structured, LLM-driven test-case generation at scale.

Defence Research and Development Organisation (DRDO), DYSL-CT
Research Intern

May – Jul 2025
Chennai, India

- Developed **PeliCAM**, a desktop framework integrating CAM, LIME and SHAP for visual analysis and interpretation of deep-learning models.
- Worked on **explainable AI (XAI)** using PyTorch, Python and PyQt; contributed to **NLP text-classification** systems for English and Indian languages.

NYU Abu Dhabi – Open Ended Lab Project
Research Collaborator

2025 – 2026
Remote

- Developed **A2QTGN**, a quantum-enhanced link-prediction model combining Adaptive Amplitude Embedding with Temporal Graph Neural Networks.
- Implemented the NewHope post-quantum key-exchange protocol as a companion cryptography project.
- Currently working on Client selection algorithms for Federated systems

Homi Bhabha Centre for Science Education (HBCSE – TIFR)
Software Intern, Learning Sciences Research Group

Feb – Dec 2025
Remote

- Contributed to an educational-cognition project analyzing 100+ students' thought processes in learning mathematical concepts.
- Co-authored "*Chaining Embodied Learning: A System to Help Develop an Integrated Understanding of Proportion, Slope and Rate of Change*", published at **ICCE 2025**.

TECHNICAL PROJECTS

Multi-Agent Test Case Generation

Tech Stack: Python, LangGraph, LangChain, GPT-4o, LLMs

Built an autonomous multi-agent **LLM** pipeline (Planner, Generator, Verifier, Judge, Supervisor) that reads raw software requirements and produces validated functional test cases with minimal human input. Replaced a manual QA workflow with an iterative, self-correcting agent loop, cutting review effort significantly.

PeliCAM – Explainable Deep Learning

Tech Stack: PyTorch, Python, PyQt, CAM, LIME, SHAP

Built a desktop framework that unifies CAM, LIME and SHAP into a single interface for auditing deep-learning models, used at DRDO to interpret classifier decisions on defence-relevant text and image data. Lets researchers visually compare which explainability method best justifies a model's prediction.

Local Pokédex – Agentic LLM System

Tech Stack: Python, LangGraph, Ollama, Local LLMs

Designed an offline agentic system where a local **LLM**-powered Fetcher agent retrieves data and a separate Verifier agent independently fact-checks it before showing results to the user, reducing hallucinated answers without relying on any cloud API.

A2QTGN – Quantum Link Prediction

Tech Stack: PyTorch, PennyLane, Temporal GNN

Proposed and implemented a quantum-enhanced link-prediction model that pairs Adaptive Amplitude Embedding with a Temporal Graph Neural Network, improving prediction of future edges in evolving graphs over classical GNN baselines. Paper submitted to a conference for peer review.

RL Pong – Deep Reinforcement Learning

Tech Stack: PyTorch, DQN, Tabular Q-Learning

Built a self-play Pong environment where Deep Q-Network and tabular Q-learning agents learn to play from scratch, using experience replay, epsilon-greedy exploration and reward shaping to stabilise and speed up convergence.

Tiger Compiler

Tech Stack: OCaml, MIPS Assembly, Compiler Design

Built an end-to-end compiler for the Tiger language from the ground up: lexer, parser, semantic analysis, type checker, IR generation, and a full backend emitting working MIPS assembly.

Chess Platform

Tech Stack: React, Node.js, PostgreSQL

Built a full-stack multiplayer chess application with real-time move syncing, server-side move validation and persistent game state, backed by a PostgreSQL database for users and match history.

X-Ray Analyser – Medical Deep Learning

Tech Stack: TensorFlow, CNN, OpenCV

Trained a convolutional neural network to flag pulmonary anomalies in chest X-rays directly from image pixels, with an OpenCV preprocessing pipeline to normalise scans of varying quality for more reliable predictions.

ACHIEVEMENTS & PUBLICATIONS

- Co-author of “Chaining Embodied Learning: A System to Help Develop an Integrated Understanding of Proportion, Slope and Rate of Change”, published at **ICCE 2025** (33rd International Conference on Computers in Education).
- **A2QTGN** – quantum-enhanced link-prediction paper submitted to a conference for peer review.
- **#8**, Inter IIT Tech Meet – 13th Edition, ISRO problem statement (ML-based lunar XRF data analysis), IIT Bombay.
- **#13**, InSpace Rocketry Competition; Ground Station Lead, ISRO / IN-SPACe / ASI.

TECHNICAL SKILLS

ML: PyTorch, TensorFlow, Scikit-learn, LangGraph, LangChain, Ollama, GPT-4o API, Deep Learning, Large Language Models, NLP, Text Classification, Explainable AI (LIME, SHAP, CAM), Graph Neural Networks, Reinforcement Learning, Multi-Agent Systems

Programming: Python, C, C++, OCaml, JavaScript, Bash

Systems & Tools: Linux, Git, OpenCV, PyQt, Compilers, Cryptography, Arduino, ESP32

LEADERSHIP & EXTRACURRICULAR

- **Sub-System Lead (Ground Station), Rocketry Club, IIT Palakkad** – Led development of ground-station control systems; secured Top 15 nationally in the InSpace rocket competition.
- **Media & Publicity Head, Petrichor 2025** – Organized technical workshops, hackathons and events for IIT Palakkad’s annual technical festival.
- **Core Team Member, The Robotics Club (TRC)** – Organized technical workshops and contributed to robotics projects.
- **NSS Head, IIT Palakkad** – Coordinated campus social service activities and community engagement events.